FRC SENSOR CAN — REVISION H REGULATED PROTOTYPE

Three fixed-threshold digital inputs. 6–22 V input; regulated 12 V and 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix-compatible device. Inspect and bench-test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Internal 5 V to 12 V converter

File: regulated_12v.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Sensor supply voltage measurements

File: rail_sense.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: fixed-threshold digital input

File: input1.kicad_sch

Input 2: fixed-threshold digital input

File: input2.kicad_sch

Input 3: fixed-threshold digital input

File: input3.kicad_sch

Original design — custom WPILib CAN protocol

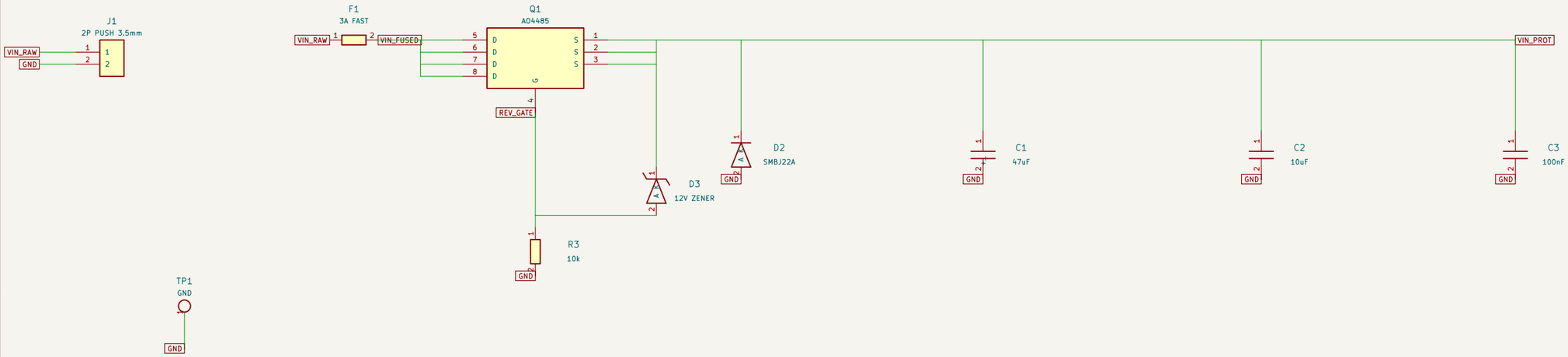
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File: FRC_Sensor_CAN.kicad_sch

Title: FRC Sensor CAN

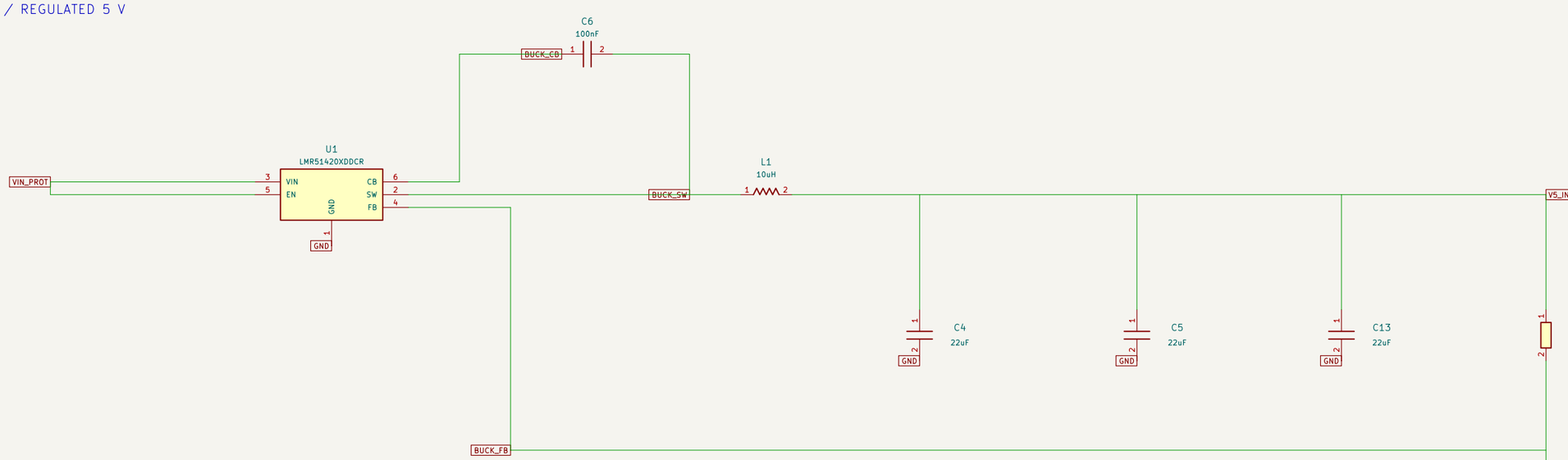
Size: A3	Date: 2026-09-09	**Rev: H**
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Battery entry and internal supplies

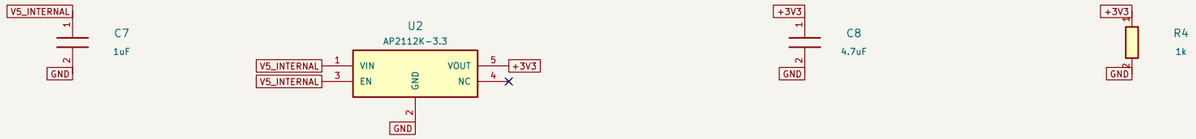
PROTECTED BATTERY ENTRY



500 kHz BUCK / REGULATED 5 V



3.3 V LOGIC SUPPLY



6–22 V input target. Single buck: 5.07 V nominal.
Protected 12 V boost supply is on the next sheet.
R1/R2 = 102k / 13.7k, 0.1%. The boost input is fed directly by this 5 V rail; U9 protects the 12 V output.
TP1 is ground; use selectors/capacitor pads for rail checks.

Prototype: bench validation required before robot use

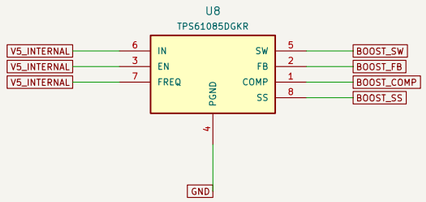
Original sensor interface design

Sheet: /Battery entry and internal supplies/
File: power.kicad_sch

Title: SC12 – Battery entry and internal supplies

Size: A3	Date: 2026-09-09	**Rev: H**
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Internal 5 V to 12 V converter















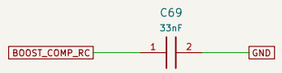














The internal converter charges only its own reservoir. U9 on the sensor-power sheet ramps sensor loads separately.
100 nF boost soft-start; 1 uF IN bypass plus 22 uF bulk. Feedback is sensed BEFORE the output eFuse.
13k / 33 nF compensation and the 3.3 uH inductor are retained. Converter output is nominally 12.003 V.

Prototype: bench validation required before robot use

Original sensor interface design

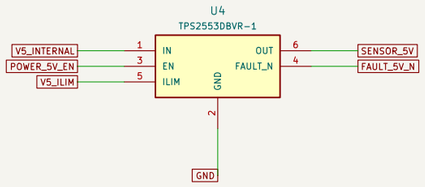
Sheet: /Internal 5 V to 12 V converter/
File: regulated_12v.kicad_sch

Title: SC12 – Internal 5 V to 12 V converter

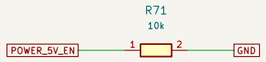
Size: A3	Date: 2026-09-09	**Rev: H**
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Protected sensor power rails

5 V BRANCH: INDEPENDENT HARDWARE LIMIT + DEFAULT-OFF ENABLE





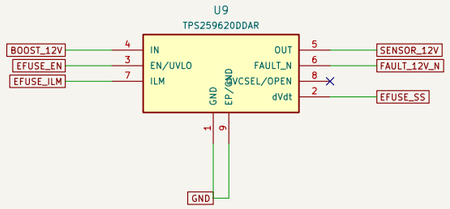






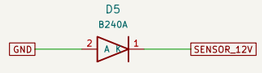
PA15 high enables 5 V. R71 holds EN low through reset and initial programming.
TPS2553-1 latches off after overload; deliberate CAN service recovery cycles EN.

12 V BRANCH: OUTPUT ISOLATION / CONTROLLED INRUSH / HARDWARE LIMIT









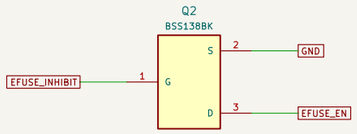




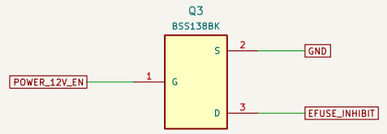














U9 is after the boost reservoir: sensor shorts do not directly discharge it.
R70 uses raw 12 V to inhibit U9 through Q2; PB7 HIGH enables via Q3. R75 defaults OFF without MCU power.
Hardware UVLO requires about 11.1 V upstream.
OVCSEL is intentionally open for the 13.8 V-class fault clamp. No reverse-powered output support.
Both rails: 300 mA useful load shared between all ports; protection never depends on firmware.

Prototype: bench validation required before robot use
Original sensor interface design

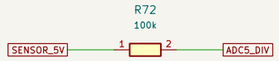
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

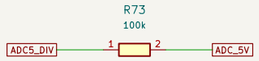
Title: SC12 – Protected sensor power rails

Size: A3	Date: 2026-09-09	**Rev: H**
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Sensor supply voltage measurements

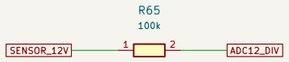
11:1 PRECISION DIVIDERS; PB0/PB1 ADC + INTERNAL VREFINT

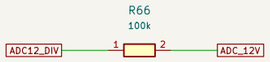


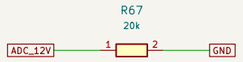




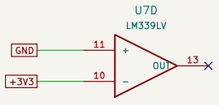












Each rail uses two series 100k 0.1% resistors and a 20k 0.1% bottom resistor.
10 nF filtering: long ADC acquisition. 30 V at the divider maps to about 2.73 V.
These dividers measure the sensor supplies AFTER protection. Firmware validates voltage and fault flags.
U7D is unused: inputs fixed to GND/+3V3 and output unconnected.

Prototype: bench validation required before robot use

Original sensor interface design

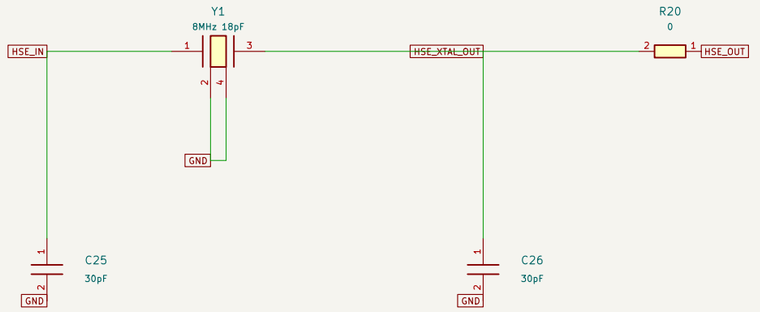
Sheet: /Sensor supply voltage measurements/
File: rail_sense.kicad_sch

Title: SC12 – Sensor supply voltage measurements

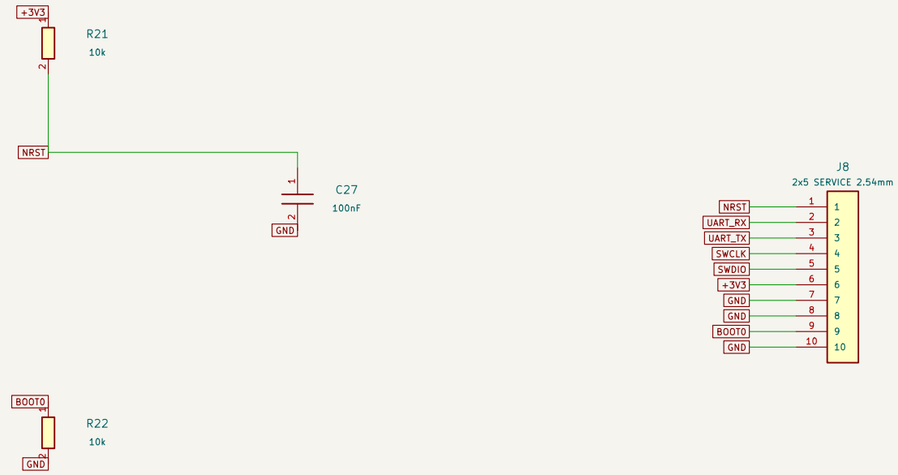
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Controller, clock, debug and status

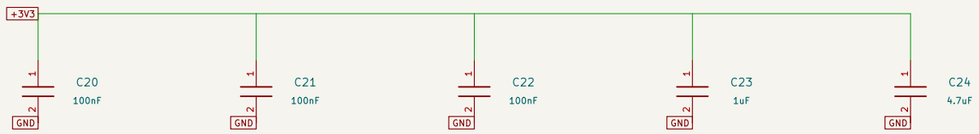
8 MHz CRYSTAL / PLL x6

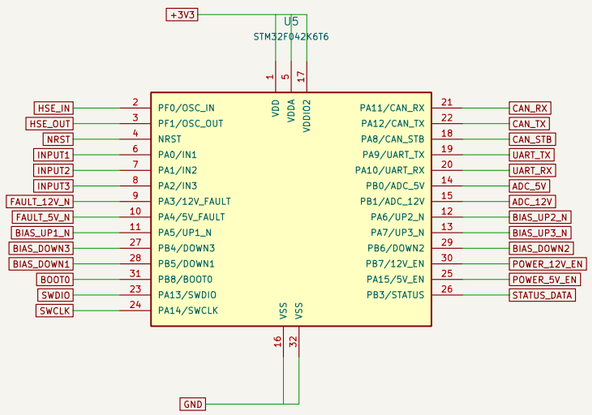


RESET / BOOT / SERVICE



LOCAL DECOUPLING





J8: hand-solder 2x5 2.54 mm header; CUSTOM electrical pinout.
Pin 6 is 3.3 V reference OUTPUT, never a programming power input.
PB0/PB1 read sensor-rail ADCs; PA15 enables 5 V and PB7 high enables 12 V.
PA3/PA4 monitor active-low fault flags. PB5/PB6/PB4 are input pull-down controls.

Prototype: bench validation required before robot use

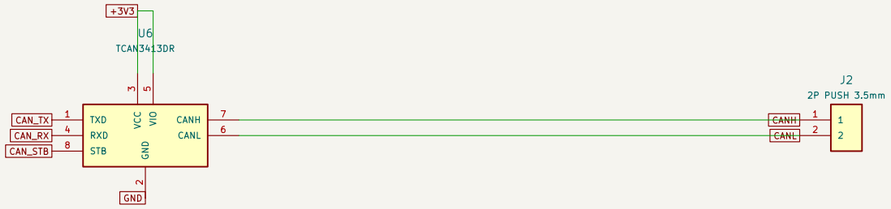
Original sensor interface design

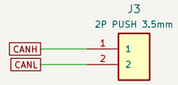
Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

Title: SC12 – Controller, clock, debug and status

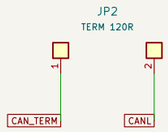
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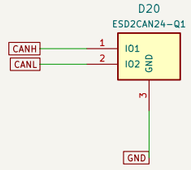
CAN physical interface and termination

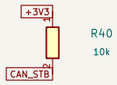


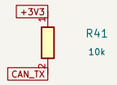










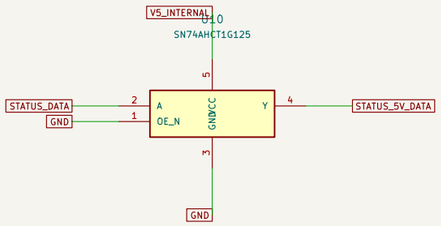




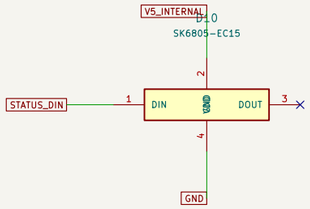


1 Mbit/s. Bridge JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

SINGLE RGB STATUS INDICATOR / INTERNAL 5 V









PB3 / TIM2_CH2 drives one SK6805-EC15 through a 5 V AHCT buffer. R31 holds data LOW on reset.
D10 remains powered when sensor supplies are disabled. DOUT is unused.
RGB patterns are firmware diagnostics, not an independent hardware power-good indicator.

Prototype: bench validation required before robot use

Original sensor interface design

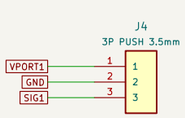
Sheet: /CAN physical interface and termination/
File: can.kicad_sch

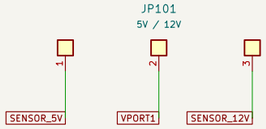
Title: SC12 — CAN physical interface and termination

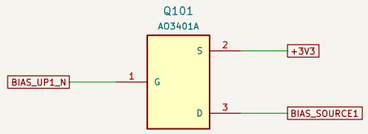
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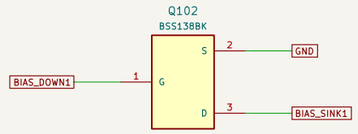
Input 1: fixed-threshold digital input

CAN-CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER

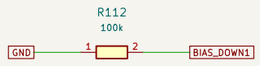


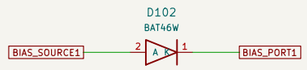


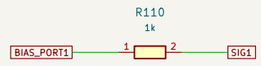


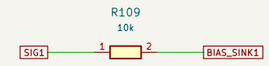




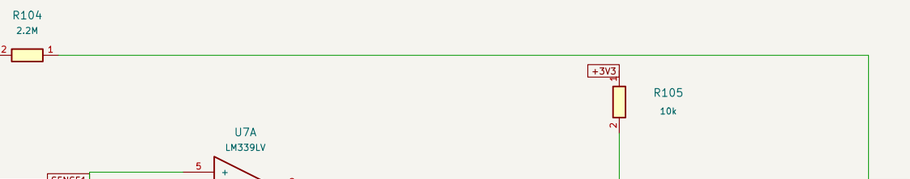


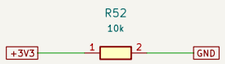






POWER: one 2 mm cap. 1-2=5 V or 2-3=12 V: no cap=off.
BIAS: 1k pull-up to internal 3.3 V through PMOS/diode; 10k pull-down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.

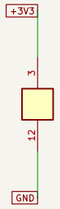














0-30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
Original sensor interface design

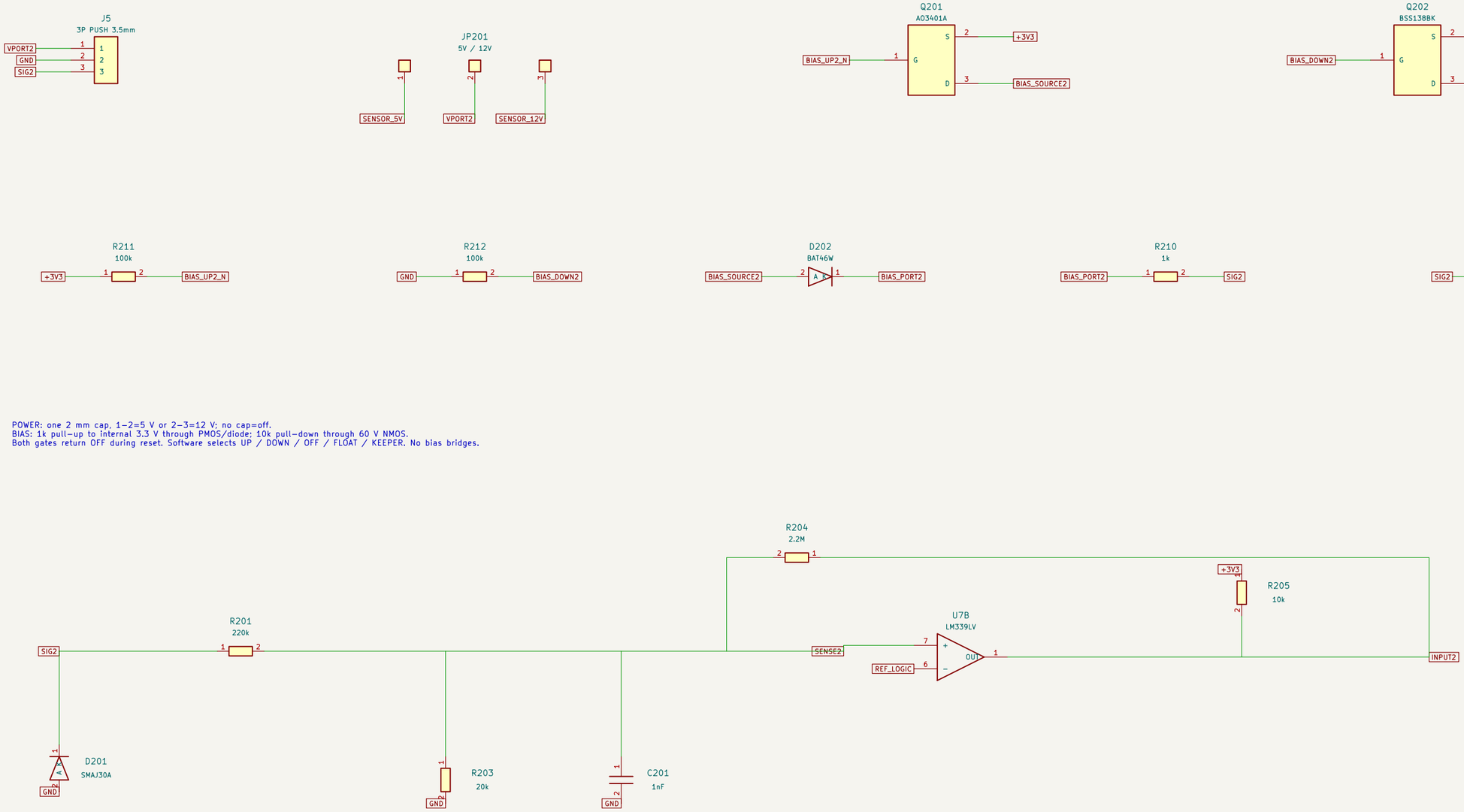
Sheet: /Input 1: fixed-threshold digital input/
File: input1.kicad_sch

Title: SC12 – Input 1: fixed-threshold digital input

Size: A3	Date: 2026-09-09	**Rev: H**
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Input 2: fixed-threshold digital input

CAN-CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER



POWER: one 2 mm cap. 1-2=5 V or 2-3=12 V: no cap=off.
BIAS: 1k pull-up to internal 3.3 V through PMOS/diode; 10k pull-down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.

The reference divider/filter and comparator power connections are on input1.
One fixed logic-compatible threshold for all three channels; no industrial threshold selector.

0-30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
Original sensor interface design

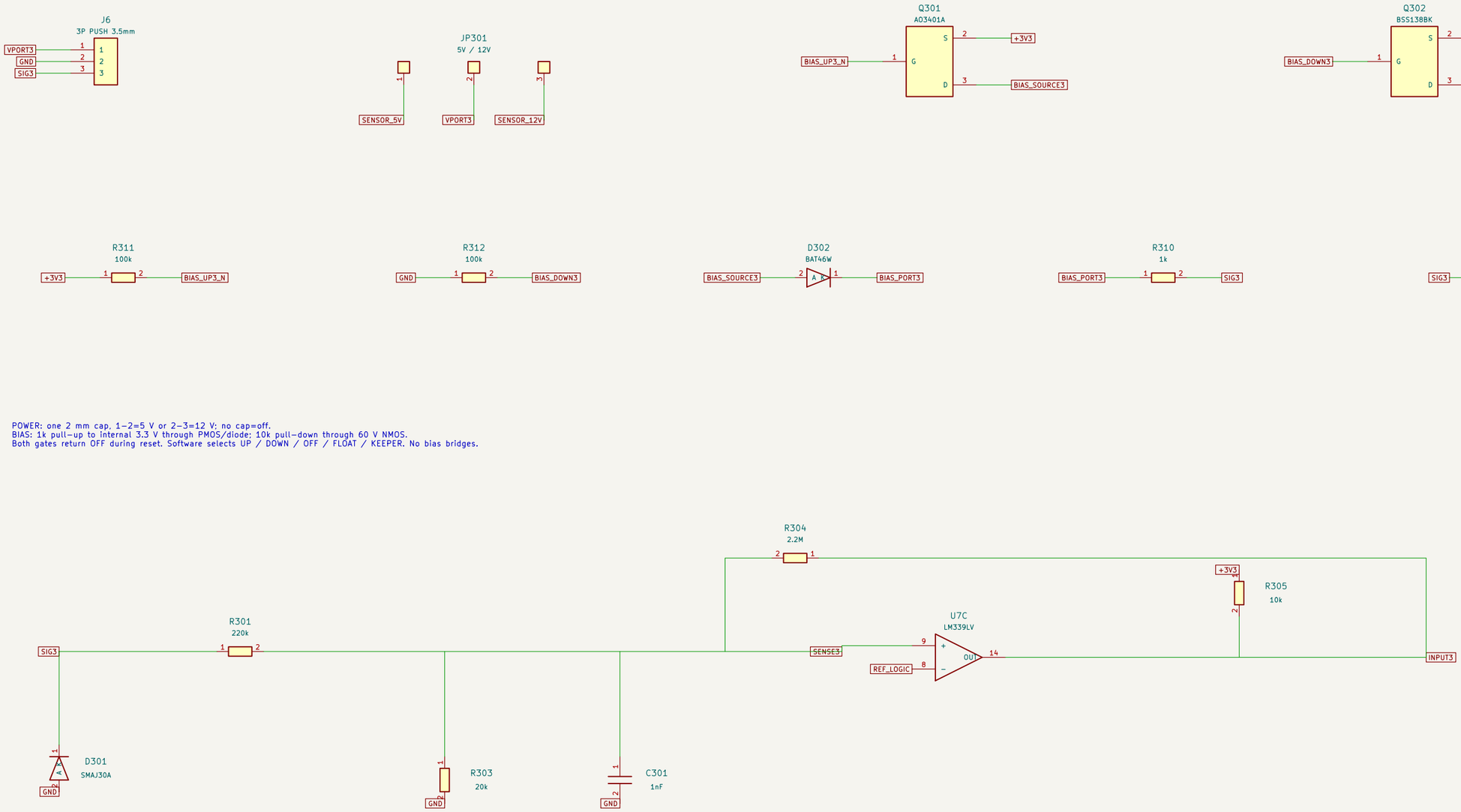
Sheet: /Input 2: fixed-threshold digital input/
File: input2.kicad_sch

Title: SC12 - Input 2: fixed-threshold digital input

Size: A3	Date: 2026-09-09	**Rev: H**
KiCad E.D.A. 10.0.6		Id: 9/10

Input 3: fixed–threshold digital input

CAN–CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER



POWER: one 2 mm cap. 1–2=5 V or 2–3=12 V: no cap=off.
BIAS: 1k pull–up to internal 3.3 V through PMOS/diode; 10k pull–down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.

The reference divider/filter and comparator power connections are on input1.
One fixed logic–compatible threshold for all three channels; no industrial threshold selector.

0–30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
Original sensor interface design

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